

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 2nd Semester B.Sc. Examination, 2022
CSE – 1201, Fundamentals of Programming

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1.
 - a) Define Program. Briefly explain about different data types in C programming language. 1+3
 - b) Write a program in C that takes minutes as input, and converts the input to hours and seconds. 4
 - c) Analyze the following program. Is it possible to execute every line of the provided code without encountering any errors? If not so, make the necessary changes to the program and display the results. 4


```
int main()
{
    int c, a=10, b;
    b=++a;
    a=b; c=a

    printf("C is %f", c);
    return 0;
}
```
 - d) Determine the value of each of the following logical expressions if
 $a = -5, b = 1, c = 5;$
 - i) $b \geq 1 \ \&\& \ c < a \ || \ b < a$
 - ii) $a < b \ \&\& \ a < c$
2.
 - a) What is the output when the following code fragments are executed? 3+3

i. <pre>int main() { int x=(int)31.5/(int)6.3; printf("%d\n",x); printf("Size=%d\n",sizeof("CSE")); printf("Lenth=%d",strlen("CSE")); }</pre>	ii. <pre>int main () { int *pc, c; c=22; printf("Address of c: %u\n", &c); printf("Value of c: %d\n",c); pc=&c; printf("Address of pointer pc: %u\n", pc); printf("Content of pointer pc: %d\n", *pc); *pc=2; printf("Address of c: %u\n", &c); printf("Value of c: %d\n",c); return 0; }</pre> // Assume the memory location is:268678
--	--
 - b) What do you mean by scope, visibility and life-time of variables? 4
 - c) Differentiate between entry controlled loop and exit controlled loop? 2
 - d) How syntax errors differ from logical errors? Which one is more difficult to find out? 2
3.
 - a) State the benefits of pointer. Show the memory representation of the following declarations; 1+3


```
int m=0, n=20;
int *ptr=&m;
n=*ptr;
```
 - b) What are the arithmetic operators that are permitted on pointers? Are the expressions $*p++$ and $++*p$ same? Justify your answer. 1+3

c) What is the output when the following code fragments are executed?

3+3

<pre>i. int main() { int x[2]={1,2}; int *ptr; ptr=x; ++(*ptr); *ptr++; x[1]--; --(*ptr); printf("(%d %d)\n", x[0]++, ++x[1]); return 0; }</pre>	<pre>ii. int main() { int x=6,y=4,z,*p1,*p2; p1=&x, p2=&y; z=x; x=y; y=z; printf("%d %d\n", *p1,*p2); *p1=(x++)+3; *p2=++x-y; printf("%d %d\n", x,y); p1=p2; printf("%d %d\n", *p1,*p2); return 0; }</pre>
--	--

4. a) Define Control Statements. Briefly describe the conditional statements with proper examples. 1+3
 b) What is Infinity Loop? Write a program to print the following sequence: 1+3
 5, 25, 125, 625, 3125, 15625
 c) Compare between **break** and **continue** with proper examples. Write a program to print all even numbers from 120 to 0 except 14 and 98. 6

5. a) Define Array with example. How to initialize one-dimensional and two-dimensional arrays? 1+2
 b) Write a C program to display the following pattern when n = 5. 5

```
*****
*****
**0**
*****
*****
```

- c) Determine and describe appropriate output of the following C program.

4

```
#include<stdio.h>
void main(){
    int i, j, n=8, a[10];
    for(i=1; i<=n; i++){
        a[i]=0;
        for(j=1; j<=i; j++){
            a[i]=a[i]+i+j;
        }
        printf("%d\n", a[i]);
    }
}
```

- d) Explain different types of errors in programming language. 2
6. a) Define file. Explain creating, opening and closing a text file in C with example. 5
 b) What is the importance of file closing? 3
 c) A file named "sorting.txt" contains a series of integer number. Write a program that read the integers from the file and after sorting these numbers in ascending order save the result in another file named "sorted.txt". 6
7. a) Distinguish between variable and pointer variable. Write a C program to print n elements stored in an array using pointer. 2+3
 b) Write a C program to take details (id, name, and mark) of n students in a structure array and show all the details. Use a structure named Student. 4
 c) Suppose you are a blood donor. Every month you donate your blood and keep record in a file. File contents are patient name, location, and date. Now write a program to create a file and store your information into that file. 5

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1st Year 2nd Semester B.Sc. Examination, 2022
CSE – 1202, Digital Logic Design

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) i. Convert the Gray code 11010011 to binary number. 2
 ii. Convert the decimal number 237.95 to BCD code. 2
 b) Implement a two input X-OR gate using only two input NOR gates. 4
 c) Simplify the expression: $Y = (A+B)(A+\overline{A}B)C + \overline{A}B + ABC$ 4
 d) Find the 9's complement of $(415.69)_{10}$. 2
2. a) Define: I) fan-out II) Noise margin III) Propagation delay 3
 b) Implement Exclusive OR gate by any Universal gate. 4
 c) Simplify the Boolean function to a minimum number of literals and draw the diagram. 4
 i) $xyz + x'z + yz + x'yz$
 ii) $(x'+y)(x+z) + xz$
 d) Simplify the Boolean function with Karnaugh map $F(x,y,z) = \sum(1,3,6,7)$ and don't care conditions $d(x,y,z) = \sum(0,2,5)$. 3
3. a) Write the output expressions of a 3-bit comparator. 3
 b) Derive the simplified output expressions for a BCD to Excess-3 code converter. 6
 c) Construct a 1x16 De-multiplexer with 1x4 De-multiplexers and describe briefly how it works. 5
4. a) What is the difference between combinational circuit and sequential circuit? 2
 b) Draw the block diagram with truth table of a 4x16 decoder using two 3x8 decoders. 4
 c) Implement the function using an 8x1 Multiplexer: 5
 $F(A,B,C,D) = \sum m(0,1,3,4,8,9,15)$
 d) Draw the circuit diagram of a full adder using half adders and OR gate. 3
5. a) Differentiate between Latch and flip-flop. 2
 b) Describe the SR flip-flop with figure. 4
 c) Implement MOD - 10 counter and explain its functions. 4
 d) Show the block diagram of IC 74193 and explain functional description of IC 74193. 4
6. a) Draw the circuit diagram of a full adder using half adders and OR gate. 3
 b) Draw the block diagram and truth table of a BCD to 7 segment display system. 4
 c) How does a priority encoder differ from an ordinary encoder? 2
 d) Implement a 3 bit Comparator circuit. 5
7. a) Design a synchronous counter that has the following counting sequence: 6
 000, 010, 101, 110
 b) Use a PROM to implement the following functions: 6
 $A(x,y,z) = \sum m(1,2,4,6)$
 $B(x,y,z) = \sum m(0,1,6,7)$
 $C(x,y,z) = \sum m(2,6)$
 $D(x,y,z) = \sum m(1,2,3,5,7)$
 c) Draw the internal logic diagram of a 32x8 ROM. 2

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PHY – 1203, Physics

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) State and explain the first and second laws of thermodynamics. 3
 b) What are Isothermal and Adiabatic processes? Establish the relation between pressure and volume in the adiabatic process. 7
 c) A quantity of air at 27°C and atmospheric pressure is suddenly compressed to half its original volume. Find the final temperature and pressure. ($\gamma = 1.4$) 4

2. a) Show that the adiabatic curves are γ times steeper than isothermal curves. 3
 b) What is Carnot cycle? Describe Carnot's cycle each stage and find out its efficiency. 7
 c) Calculate the molecular kinetic energy of 1 gm of hydrogen gas at 50°C, given that the molecular weight of Hydrogen is 2 and $R = 8.3 \times 10^7$ ergs/gm-mol. 4

3. a) Define simple harmonic motion and explain its characteristics. 3
 b) Derive an expression for the differential equation of motion of a body executing simple harmonic motion and find out the solution of it. 7
 c) Two simple harmonic motions acting simultaneously on a particle are given by the equations $y_1 = 2 \sin(\omega t + \frac{\pi}{6})$ and $y_2 = 3 \sin(\omega t + \frac{\pi}{3})$. Calculate (i) amplitude and (ii) phase constant of the resultant vibration. 4

4. a) Define crystal lattice, unit cell, and coordination number. 3
 b) Show that for cubic crystal structure the interplanar distance, $d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$. 7
 c) Sketch the following planes of a cubic unit cell: (011), (0 $\bar{1}$ 0), (110), ($\bar{1}$ 11) 4

5. a) Define: (i) Bravais Lattice (ii) Primitive Cell (iii) Miller index 3
 b) Show that for bcc and fcc crystal structures the lattice constants are given by $a_{bcc} = \frac{4r}{\sqrt{3}}$ and $a_{fcc} = \frac{4r}{\sqrt{2}}$. 7
 c) In a crystal, a lattice plane cuts intercepts of a, 2b, 3c along the three axes, where a, b and c are primitive vectors of the unit cell. Determine the Miller indices of the given plane. 4

6. a) Explain the terms 'Phase velocity' and 'Group velocity'. 4
 b) Obtain an expression for the differential equation of a plane progressive wave. 6
 c) Verify that the harmonic wave function $\Psi = A \sin(kx - \omega t)$ is a solution of the one-dimensional differential wave equation. 4

7. a) Define any three: (i) Equipartition of energy (ii) Entropy (iii) Coherent source (iii) damped harmonic oscillation (iv) Huygen's principle (v) double refraction. 3
 b) What is Brewster's law? From Brewster's law show that the reflected and refracted rays are perpendicular to each other. 7
 c) A mass of gas at 80°C and 50 atmospheric pressure is adiabatically expanded to 10 times its initial volume. If the ratio of specific heats is 1.4. Calculate the resulting pressure and temperature. 4

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MATH – 1204, Method of integration, differential equations and Series

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) A curve C is defined by the parametric equation $x = t^2$, $y = t^3 - 3t$ 8
 (i) Show that C has two tangents at the point (3,0) and find their equations.
 (ii) Find the points on C where the tangent is horizontal or vertical.
 (iii) Sketch the curve.
 b) Solve the initial value problem for $\frac{dy}{dx} + y = x$, $y(0) = 4$. 6
2. a) Test the Convergence of the following series (any two). 8
 (i) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n+1} + \sqrt{n}}$
 (ii) $\sum_{n=1}^{\infty} \frac{1.3.5.7 \dots (2n-1)}{n^4}$
 (iii) $\sum_{n=1}^{\infty} \frac{4.7.10 \dots (3n+1)}{1.2.3 \dots n} x^n$
 b) A bottle of orange juice being taken out of refrigeration at 2^oc warm up 5^oc in 5 minutes while sitting in a room of temp 23^oc. How warm will the orange juice be if left out for 15 minutes? 6
3. a) Find the quadratic approximation of $f(x) = \cos x$ for x near 0. 4
 b) If the ellipse $x^2 + 4y^2 = 16$ is rotated about its minor axis, the resulting ellipsoid is called an oblate spheroid. Find the surface area of this ellipsoid. 5
 c) For the following population model (in billions)
 $\frac{dP}{dt} = P - 144P^2$, $P(0)=7$ 5
 Describe the behavior of $P(t)$ as $t \rightarrow +\infty$.
4. a) Use Taylor's series method to solve 7
 $\frac{dy}{dx} = x + y$
 with $y(1) = 0$, numerically up to $x = 1.2$ with $h = 0.1$
 b) Given, 7
 $\frac{dy}{dx} = \frac{y-x}{y+x}$; $y(0) = 1$
 Find y for $x = 0.1$ by Euler's method.
5. a) Define order and degree of a differential equation 2
 b) Solve the following differential equation: 3
 $\frac{dy}{dx} = (4x + y + 1)^2$
 c) Define homogeneous differential equation. Solve: 2+7
 $\frac{dy}{dx} = \frac{y}{x} + \cos \frac{y}{x}$
6. a) Evaluate: 3 × 3
 I) $\int_0^{\frac{\pi}{2}} \cos^3 x (\sin x)^{\frac{1}{4}} dx$ II) $\int_{-\infty}^{\infty} \frac{x}{x^2+1} dx$ III) $\int_0^{\infty} x e^{-x^2} dx$
 b) Evaluate $\int_0^{\pi} \sin x dx$ by dividing the interval into 8(eight) strips using Simpson's rule. 5
7. a) Find the area of the region that lies inside the circle $r = 3\sin\theta$ and outside the cardioid $r = 1 + \sin\theta$. 5
 b) Find the length of one arch of the cycloid $x = r(\theta - \sin\theta)$, $y = r(1 - \cos\theta)$. 5
 c) Find the Maclaurin Series expansion of $f(x) = e^x$ 4

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ENG – 1205, Developing English Language Skills

Total Marks: 70

Time: 3 Hours

Part A (Comprehension: 15 marks)

1. Read the following passage and answer the questions given below:

The internationalization of higher education stands as an intricate phenomenon that reflects the profound transformation of academic landscapes in an increasingly interconnected world. At its core, this paradigm denotes the transcendence of traditional geographic and cultural boundaries in the pursuit of knowledge and intellectual enrichment. Universities and colleges across the globe are reorienting their strategies to not only welcome a diverse spectrum of students and scholars from around the world but also to establish international partnerships and collaborative research initiatives that bolster the cross-pollination of ideas. This seismic shift towards internationalization underscores the recognition that addressing the complex challenges of the modern era, such as climate change, global health crises, and socio-economic disparities, requires a synthesis of diverse perspectives and expertise. Moreover, the internationalization of higher education is fostering an enriching tapestry of multiculturalism, promoting cross-cultural understanding, and preparing students to navigate the complex globalized landscapes of the 21st century.

However, as the trajectory of internationalization unfolds, it is not devoid of challenges. Educational institutions must contend with issues such as equitable access to global educational opportunities and the maintenance of academic quality in a diverse learning environment. The promotion of linguistic and cultural diversity is a key facet of internationalization, but it necessitates addressing language barriers and ensuring that students from various linguistic backgrounds can fully engage in academic discourse. Additionally, there is a growing need for institutional and governmental policies that facilitate the integration of international students, thus optimizing their educational experience. Furthermore, the internationalization of higher education is inextricably linked to concerns regarding the commodification of knowledge, as the pursuit of profit can sometimes undermine the broader educational mission. In navigating these complexities, higher education institutions must adopt a nuanced approach that embraces the manifold benefits of internationalization while vigilantly addressing its inherent challenges.

- a) Choose the correct answer:

5

- i. What is the primary impact of educational internationalization?
a) Pedagogical tradition b) Academic homogeneity c) Paradigm alteration d) Globalisation
- ii. When considering the challenges of internationalization, what issue significantly perturbs academic institutions?
a) Profit maximization b) Linguistic diversity c) Equitable access d) Disciplinary synergy
- iii. What is the relationship between the push for linguistic and cultural diversity and language barriers?
a) Synergy b) Irrelevance c) Antagonism d) Concordance
- iv. How can institutions enhance the integration of international students?
a) Pedagogical isolation b) Cultivating xenophobia c) Fostering intercultural interactions d) Encouraging insularity
- v. In the context of the commercialization of knowledge, what is the most judicious approach for addressing concerns?
a) Unbridled capitalism b) Pragmatic acknowledgement c) Profit maximization d) Academic isolation

- b) Answer the following questions:

5

- i. What's the core concept of internationalization in education, and how does it challenge traditional boundaries?
- ii. What are the main challenges faced by institutions aiming for internationalization, particularly in terms of access and quality?
- iii. How does the push for linguistic and cultural diversity intersect with language barriers in global education?
- iv. How can international student integration be improved for an optimal learning experience?
- v. What are the concerns related to the commercialization of knowledge in the context of internationalization, and how can they be addressed?

- c) Write a summary of the passage.

5

Part B (Grammar: 25 marks)

2. Fill in the gaps with the correct form of verbs: 5
- Parents (a) ____ (be) the blessings of God to us. When a child (b) ____ (bear), their joys (c) ____ (know) no bounds. They (d) ____ (start) (e) ____ (think) of (f) ____ (nurse) and (g) ____ (bring) up their children. They (h) ____ (remain) safe in the custody of their parents since birth. Parents never mind (i) ____ (take) pains for the upbringing of their off-spring. We never (j) ____ (offend) our parents.
3. Transform any five (05) of the following sentences as directed. 5
- a) The weather was stormy, but we decided to go hiking. (complex)
b) While the team had a talented offense, their defense struggled to keep up. (compound)
c) She had been studying diligently, and she aced the exam. (simple)
d) Despite her strong desire to attend the concert, she couldn't secure any tickets. (compound)
e) The rain was pouring, and the wind was howling through the trees. (simple)
f) Sarah wants to study medicine, and John plans to become an engineer. (complex)
4. Use appropriate prepositions in the blanks: 5
- Patriotism is the quality inherent (a) ____ man. He feels a natural attachment (b) ____ his motherland. It is essential (c) ____ the progress of a country. The patriotic people are different (d) ____ other people. Their memories don't sink (e) ____ oblivion. They are worthy (f) ____ praise. They lay (g) ____ their lives for the good of the country. They are aware (h) ____ their position in the society. They are celebrated all (i) ____ the world. We are indebted (j) ____ them.
5. Correct any five (5) of the following sentences: 5
- a) It is she but not I am guilty.
b) Do you know who am I?
c) The boy is too strong to carry the heavy load.
d) Had I the wings of a bird, I would have meet you.
e) It is high time we should check environment pollution.
f) She is reading in the same class for three years.
g) No sooner he had gone to college, the bell rang.
6. Fill in the gaps with appropriate articles (a/an/the). Put a cross (x) if it requires. 5
- Industry is (a)---key to success in our life. (b)---industrious can prosper in life. (c)--- man who does not undertake (d)--- hard work can never go (e)---long way in (f)--- world. Many (g)---man is not aware of (h)--- significance of (i)---diligence for which. they cannot reap (j)---benefit of success.

Part C: Writing (30 marks)

7. Write an application to the Managing Director of Rangs Electronics Bangladesh for the post of a Computer Engineer along with a complete CV. 8
8. Write an e-mail to your friend condoling on his/her father's death. 7
9. Write a dialogue between two friends for strengthening cyber security in our country. 7
10. Write a paragraph on "4th Industrial Revolution" or "Bad Effects of Unplanned Urbanization". 8